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The mean and turbulent flowfields in a model RQL gas-turbine combustor

P. Anacleto, M. V. Heitor, A. L. N. Moreira

Abstract This paper presents a study of the flow patterns typical of "Rich-burn, Quick-quench, Lean burn", RQL, gas-turbine combustors based on detailed velocity measurements of the flow in a water model of a can-type combustor.

The flow split along the model combustor has been adequately established in order to simulate the real combusting flows. A laser-Doppler velocimeter has been used to analyse the development of the axial, radial and tangential, mean and turbulent velocity components along the combustor and the results are provided for the rich, quench and lean zones. The analysis includes consideration of four different combustor configurations, which allowed to perform a parametric study in terms of the flow boundary conditions and flow split. The intensity of the primary recirculation zone, the penetration of the dilution jets and the uniformity and swirl level of the flow at the outlet of the model are studied for the various configurations, and the related implications for the design of practical combustors are discussed.

1 Introduction

Rich-burn, Quick-quench and Lean-burn (hereinafter referred to as RQL) has been considered as a suitable technology to decrease the production of oxides of nitrogen in conventional gas-turbine combustors, e.g. Bahr (1995), and was one of the technologies proposed as a result of the pilot phase of the European "Low Emission Combustor Technology Program", in order to reduce pollutant emissions in future aircraft engines (Joos and Pellischeck 1995). This technique consists in burning the fuel far above the stoichiometric conditions in a first stage of the combustor (i.e., the rich zone) to avoid the occurrence of high temperatures responsible for the production of thermal NO_X. The unburned hydrocarbons, the CO and the carbon generated in the rich zone will burn at lean conditions and low temperatures in a third stage of the combustor (i.e., the lean zone), for which a sufficiently long residence time is required. The transition from fuel-rich to fuel-lean combustion is obtained through a rapid mixing of the rich mixture with dilution air jets at the second stage of the combustor (i.e., the quench zone), where the flow is accelerated in order to limit the residence time in potential stoichiometric regions.

The development of this type of combustor represents a major challenge (e.g. Bahr 1995) and must be planned with the following objectives: i) to improve wall cooling in the primary zone, where radiation from the carbon particles is very intense but any wall air injection should be prevented to avoid the formation of stoichiometric regions; ii) to optimise the mixing pattern downstream of the quench zone, which is strongly influenced by the effectiveness of the dilution jet penetration at the quench stage; and iii) to limit the carbon production at the rich conditions occurring in the primary zone and to improve its combustion in the lean zone. Although a considerable progress has already been made in the development of this type of low-NO_X combustors, they are inherently more complex than current technology combustors and require additional and extended research. In particular, the solution of the technical problems mentioned before demands a detailed understanding of the internal flow structure in the new combustors. However, measurements of the complete velocity field in real combustors under burning conditions are extremely difficult, and seriously limited by the geometry of the combustor, time and the number of instruments and skills required. These limitations have attracted the use of isothermal, constant density systems (Heitor 1989), which, above all, provide the basis for the implementation of the boundary conditions and the development of the turbulence models required in the calculation of practical flow fields. For these constant density experiments, the use of a water flow in a Plexiglas model is often preferred due to the possibility of optical access and the easy of laser-Doppler velocity measurements in water flows (e.g., Jones and Whitelaw 1985).

In this context, water flow models have been used in the past, which provided improved understanding of the complex flow pattern in gas turbines combustors (e.g. Green and Whitelaw 1983; Koutmos and McGuirk 1989; Bicen et al. 1989, McGuirk and Palma 1995). However, it is known that exact similarity

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between isothermal and reacting flows is clearly not possible. For example, the primary recirculation zone used in the actual combustors to stabilise the flame is shorter and wider than that in the equivalent isothermal flow; the mass flow recirculated is smaller and the reverse velocities larger due to the decrease in density; the velocity fluctuations are generally larger and the local turbulence intensities smaller owing to thermal expansion (e.g. Heitor and Whitelaw 1986). A primary measure applied to gas turbine combustor technology is to use similar flow splits along the combustor, namely through independent control of primary and secondary flows (e.g. Bicen et al. 1989). Nevertheless, it is clear that apart from the details which cannot be conveniently studied in isothermal non-reacting models, a water flow rig may provide the necessary insight into gas-turbine flows to allow to improve the design of practical combustors, namely through the evaluation of the turbulent models used in the numerical simulation of related flows (e.g. Coupland and Priddin 1987; Jones and Whitelaw 1985; Jones and Kakhi 1995).

The work reported here is aimed at contributing for these objectives and includes the analysis of the flow in a Plexiglas water model of a practical RQL combustor (see Turrini 1994, for details), which was scaled at 1 : 1, keeping similar flow splits. The work includes the use of a laser-Doppler velocimeter to study the influence of the various parameters which govern the performance of an RQL combustor, such as the swirl level in the rich zone, the quench module geometry, and the momentum of the dilution jets and of the annular impinging jet.

The paper is organised as follows: the model combustor and the flow conditions considered throughout the work are given in the next section; the velocimeter is described in Sect. 3, together with assessments of the accuracy of the measurements and the method used to determine the flow split along the combustor. The results are then presented and discussed in Sect. 4 and the main conclusions are summarised in Sect. 5.

2 The model combustor

2.1 The combustor geometry

The experiments were performed in a Plexiglas combustor model placed in a long horizontal water tunnel, as shown in Fig. 1. The facility includes a recirculating system whereby water is pumped from a constant-head discharge tank to a large pressurised tank (up to 2 bar). The water is supplied to the model through an inlet section with straightners and screens, located about 1.5 m downstream of the pressurised tank.

Figure 2a shows the layout of the overall RQL test model. The architecture of the model consists of three different modules (namely, rich, quench and lean) assembled concentrically inside a larger peripheric can with 125 mm in diameter and 400 mm in length, so that a surrounding annular water flow is formed to feed the liner inlet ports accordingly to the related pressure drops. The primary flow enters in the model via an inner stainless steel pipe with 40 mm inside diameter, which provides an orifice meter and a valve to measure and regulate the primary/secondary flow ratio. The primary flow is then directed to the head of the combustor through a diverging pipe. A coaxial swirler, similar to that used in the real combustor, is fitted around the fuel atomizer at the head of the model. For the results reported here the fuel atomizer was blocked.

The rich module consists of a cylindrical barrel of 80 mm internal diameter and 62 mm length. The quench module is located just downstream for the rich module and consists of a 20 mm length truncated cone, which provides a 5 : 8 area contraction, followed by a 20 mm length constant section of 63 mm inside diameter with eight holes disposed symmetrically along the wall. Part of the secondary stream is diverted through these holes providing jets emerging normally to the direction of the primary stream to simulate the quench jets in practical combustors. Two different quench modules were used in the present work with a similar geometry, but with circular quench holes with 6.2 mm in diameter, Fig. 2b, or, alternatively, with oblong holes with 11.3 mm in length and 2.8 mm in width and placed at 42° with the cord formed by their centres, Fig. 2c.

At the junction of the rich with the quench modules, there is an annular passage with 8 mm in width. Part of the secondary stream is diverted into the primary flow through this passage to simulate the wall-cooling jet in the practical combustor, which flows along the contraction wall inclined at 42° with the direction of the primary stream. This part of the secondary flow is controlled through the positioning of a perspex ring, which can slide forward and backward to allow to change the blockage at the inlet port of the annular cooling jet.

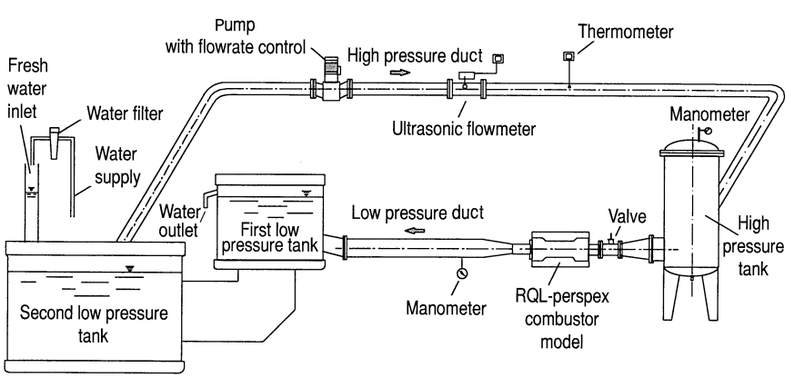


Fig. 1. Schematic drawing of the experimental facility

The third module, which simulates the lean zone in practical combustors, includes a 98 mm diameter cylinder with 76 mm in length and is followed by a sudden enlargement to 124 mm with 45 mm in length. At the expansion between the quench and the lean zones, 96 small holes with 0.8 mm in diameter discharge part of the secondary fluid into the primary flow, which simulates the air used to cool the wall in this region. In one of the configurations considered here, four additional holes with 8 mm in diameter were included in the lean zone, to provide crossing jets to decrease the swirl motion of the outlet flow. Nevertheless, all the configurations tested include also 60 concentric holes with 4.1 mm in diameter in the plan of the sudden expansion downstream of the lean zone, which simulate the dilution jets in the practical combustors.

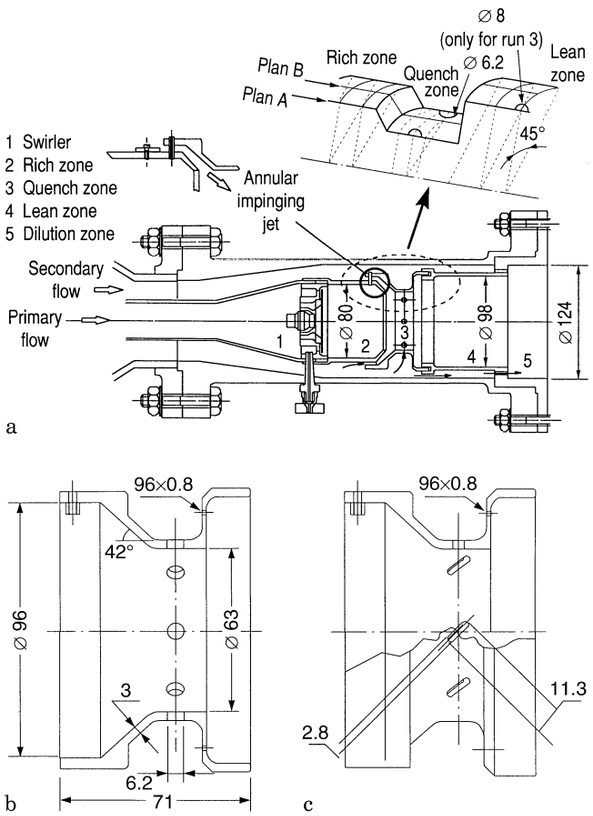


Fig. 2a–c. Schematic drawing of the combustor model (source: Turrini 1994). **a** The RQL water test model and measuring stations along the combustion model, with indication of two typical measuring plans (A and B, respectively across the quench holes and between two holes); **b** quench model with circular holes; **c** quench model with oblong hole

2.2
The flow conditions

The measurements were performed at a water temperature in the range $15 < T\ (^\circ C) < 19$ and the overall flow rate was kept constant and equal to 7.5 kg/s. The Reynolds number based on the mean axial velocity at the outlet section varied between 39×10^5 and 64×10^5.

Table 1 shows the various flow conditions considered, which have been planned to study the effects of the quench jets, of the flow split between the primary and the secondary flow and of the cooling jet at the inlet of the quench module, on the flow pattern typical of RQL combustors.

3
The experimental procedure and the measuring accuracy

3.1
The experimental technique

The mean and fluctuating velocities were measured by a laser-Doppler velocimeter based on a 2W Argon-lon laser and operated in the dual-beam, forward-scatter mode with sensitivity to the flow direction provided by light-frequency shifting from acousto-optic modulation (Bragg cells), normally set at 2 MHz.

The light scattered by naturally-occurring centres in the water was collected by a lens (focal length of 200 mm) and focused onto the pinhole aperture (0.3 mm) of a photomultiplier (OEI, type LD-0-810) with a magnification of 1.0. The output of the photomultiplier was band-pass filtered and the resulting signal processed by a commercial frequency counter

Table 1. Flow conditions

Run	Main geometric features	m_t (kg/s)	$Re^* \times 10^5$	T (°C)	P_s (bar)	U_0 (m/s)
1	Circular quench holes	7.5	64	19 ± 1	0.85	0.621
2	Oblong quench holes	7.5	64	16 ± 1	0.85	0.621
3	Similar to Run 1, but with a different flow split and 4 new dilution holes ($d = 8$ mm), in the lean zone	7.5	57	15 ± 1	0.65	0.621
4	Similr to Run 1, but without the impinging, cooling annular jet	7.5	39	18 ± 1	0.95	0.621

$^*Re$ – Reynolds number, $Re = (4 \times m_t \times (1 - \lambda_d))/(\pi \times \mu \times d_0)$
m_t – total mass flow rate (kg/s)
μ – dynamic viscosity
d_0 – diameter at the outlet of the primary flow in the water model
λ_d – (dilution flow rate)/total flow rate
ρ – specific weight
U_0 – bulk velocity at the exit section of the lean zone
P_s – state pressure at the upstream duct
D_0 – inside diameter of the peripheric can (125 mm)

(TSI 1980-B), which was operated in the single-measurement-per-burst mode with fixed cycles ($N=8$) validated in the time domain within 1%. The validated data were transferred to a laboratory 16-bit minicomputer, where the time averaged moments of velocity were evaluated.

The complete LDV system was mounted on a three-dimensional traversing unit, allowing the positioning of the laser-velocimeter control volume within ± 0.25 mm. Measurements were obtained with the laser beams in the horizontal and vertical planes and by traversing the control volume along the horizontal and vertical directions to allow the determination of the axial, U, radial V, and tangential, W, mean and turbulent velocities. The results are presented either along a longitudinal plan containing the quench holes (i.e., plan A in Fig. 2a), or along a longitudinal plan passing between a pair of holes (i.e., plan B in Fig. 2a).

A cylindrical polar system was used to represent the results, with the origin of the radial coordinate taken at the centreline of the combustor and the origin of the axial coordinate taken at the entrance plane of the rich zone. The tangential velocity is taken positive in the anticlockwise direction, as viewed from upstream.

The effects of refraction at the external surface were reduced by immersing the model in a water through. The location of the measuring volume and the associated changes in the transfer function were calculated as suggested by Bicen (1982) and Pereira (1989). Measurements of the axial, U, and tangential, W, velocity components along the horizontal diameters may be obtained without refraction corrections (which were typically found to be below 1%). A limitation lay in the measurement of the radial velocity, which becomes progressively more difficult as the measuring volume approaches the top and bottom pipe walls, because of the increasing refraction and dispersion of light at the solid–liquid interfaces. The furthest position at which measurements can be reliably obtained is about 0.8 (r/R). The deterioration in the signal quality of radial velocity as the measuring volume lies further off axis, impairs the reliability of demodulation and measurements were avoided when the data-rate decreases below 50% of the data rate at the centreline. However, no corrections have been applied to the positioning of the measuring point and to the transfer function of the velocimeter. A further restriction imposed by the curved pipe surface is that quantification of the cross correlation $\overline{u''v''}$ is unpracticable with the present LDV system.

3.2
The accuracy of the velocity measurements

This section summarises the main sources of error related with the confined water flow and with the optical arrangement used and gives estimates of those errors.

The most serious potential systematic errors may be due to measuring-volume positioning, Doppler-frequency broadening and velocity bias. The main sources of random errors affecting the accuracy of the measurements are due to statistical sampling uncertainty.

The uncertainty in locating the probe volume at the centreline of the combustor is estimated to be less than half of the effective probe volume length and, therefore, below 0.5 mm. This uncertainty varies as the beams traverse the curved pipes because of the wall curvature effects, but may be estimated from the relations given by Bicen (1982). These relations depend on the velocity component of interest and on the shape of the surfaces at the measurement station and show that the error incurred in the half angle between the beams in water is negligible. Similar calculations show that the positioning uncertainty is only significant very close to the pipe walls. The maximum systematic errors are typically found to be about 1% of the undisturbed value for the measurements of the axial and tangential velocities, and up to 5% for the radial velocity, but measurements close to the wall have been excluded as mentioned before.

Doppler broadening occurs because the Doppler-frequency probability density function is affected by sources other than true velocity fluctuations at a point. Full accounts of these sources are to be found in the literature (e.g. Durst et al. 1981), and are not given here. The net effect is that the measured distribution is broader than that of the velocity. However, the main errors incurred in the measurements are due to transit broadening, Zhang and Wu (1987), and for the present tests are below $5 \times 10^{-3}\ U_{max}$, where U_{max} is the maximum velocity at each radial profile.

Although various weighting methods have been proposed to correct for velocity bias effects (e.g. Durst et al. 1981), no corrections were applied to the measurements reported here. The systematic errors that could have arisen were minimised by using high data acquisition rates in relation to the fundamental velocity fluctuation rate, as suggested for example by Dimotakis (1978) and Erdman and Tropea (1981). This could be easily achieved because the rate of naturally-occurring particles was sufficiently high for the flow conditions considered here.

Statistical sampling uncertainty due to the finite size of the sample was estimated based on the analysis of Yanta and Smith (1978). The number of individual velocity values used to form the averages were always above 10 000 so that, for a 95% confidence interval, the largest statistical errors are smaller than 0.5% and 3%, for the mean and variance values respectively.

Spectral analysis of LDV signals was carried out by sampling and holding the velocity at each new Doppler burst until another valid signals arrives. The condition under which the output of the counter yields a satisfactory spectral analysis has been discussed by, for example, Edwards and Jensen (1983), Adrian and Yao (1985) and Srikantaiah and Coleman (1985) and was satisfied in the present experiments. A mean sampling frequency 2 KHz was used, suggesting that structures of the order of 1.0 mm and above can be resolved. This is at least one order of magnitude smaller than the expected large length scales in the flow, which implies that the energy containing motions are resolved by the counter.

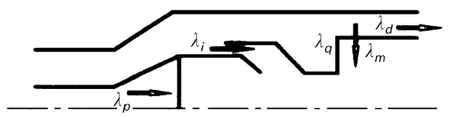
3.3
The flow split analysis

A primary measure to guide the use of water models in the study of gas-turbine combustors is to establish adequate flow splits (e.g. Bicen et al. 1989). This has been achieved in the present work through a detailed analysis of the secondary flow, as described below.

A proper method to define the flow split between the primary and the secondary streams was found based on the

Table 2. Flow split along the combustor

Run	λ_p (%) (flowmeter)	λ_l (%)	λ_q (%)	λ_m (%)	λ_d (%)
1	24	29	10	—	27
2	24	38	11	—	27
3	15	31	13	8	33
4	25	0.0	18	—	57



rate of the primary flow (measured by the orifice flowmeter) and on the total flow, measured by an ultrasonic meter. The orifice meter has been calibrated in the present rig to account for sensitivity to the flow conditions at the inlet section. The calibration was performed against the ultrasonic flowmeter, with an error smaller than 1%, and the total error of this method is considered to be smaller than 5%.

The flow split along the combustor was obtained based on the volumetric rate of water flowing through the secondary stream. This was calculated at several axial locations by integration of radial velocity profiles, based on the method described by Spencer et al. (1995) for pipe flows, which includes the use of the law of the wall in the regions where LDV measurements could not be obtained (i.e. for $r/R > 0.95$).

The analysis has shown that the secondary flow is axisymmetric only downstream of the quench module, which may be due to the perturbation induced by the three supporting screws located at the dome of the water model. As a consequence, an accurate determination of the flow rate required consideration of the angular variation of mean velocity in the secondary (i.e. outer) flow. The procedure consisted in the analysis of four radial profiles along half the distance between two consecutive screws. Since the mean flow has been shown to have an angular repeatability in each sector between two consecutive screws, these profiles are effectively representative of 24 radial profiles around the secondary flow.

In the lean zone, the secondary flow was found axisymmetric and integration of the velocity profiles was used to obtain the dilution flow split. The related error in the flow measurement has been estimated to be smaller than 5%. Nevertheless, larger errors may arise due to the fact that the flow is not fully developed and of the inadequacy of the 'law of the wall'.

Table 2 summarises the flow split along the combustor for all the flow conditions considered in the present experiments, as percentages of the total flow rate.

4
Results and discussion

The results are presented and discussed in this section under two headings. First, Run 1 is considered as a standard case, for which the main features of the flow are described in detail. Secondary, a parametric study is described based on the analysis of the effects of the shape of the quench holes, of the flow split at the entrance of the combustor and of the aperture of the annular cooling jet, on the flow pattern in the model combustor.

To ease analysis, the results are presented in terms of velocity vectors along the longitudinal planes A and B of Fig. 2a) and/or along transversal sections of the combustor. The former have been calculated from the measurements of the axial and radial velocity components, while the latter represent the tangential velocity component.

4.1
The standard case – RUN 1

Figure 3 shows sample results characteristic of the mean flow along the combustor. The primary zone is characterised by a central swirl-driven recirculation zone, which is essential to provide mixing and stable combustion, Mellor (1990). The maximum measured reverse velocity is about 0.98 U_0 and, therefore, 27% of the maximum axial velocity for larger radius. In addition, the maximum tangential velocity is about 2.6 U_0 and occurs in the shear layer surrounding the central reverse flow zone, in a way that the central core of the flow rotates as a solid body. On the other hand, this tangential mean velocity distribution remains constant in the outer flow region, suggesting a linear decrease of the angular momentum with the radial distance. In this zone, the results show an annular recirculation zone established at the inlet of the sudden expansion, whose size may depend on the geometry of the dome and also on the size of the inner recirculation zone as discussed, for example, by Rhode et al. (1983). For the present geometry and flow conditions, the length of that annular recirculation zone is limited by the contraction at the entrance of the quench zone.

In the secondary stream, an annular recirculation zone is formed at the expansion close to the entrance of the quench holes, which promotes the uniformity of the momentum fluxes in all the jets. The internal flow in the quench module has a cyclic repeatability comprising eight similar sectors of 45°. Measurements were obtained along the two longitudinal planes of Fig. 2a), respectively passing through the centre of the quench holes (in plane A), and passing midway between two consecutive holes (i.e. plane B). The pattern followed by the quench jets is similar to those represented by jets in cross flow (e.g.: Andreopoulos and Rodi 1984), but with a weak penetration. This may be partially due to the deflection imparted by the impingement of the annular wall-cooling jet, which also influences the central swirling flow. In fact, an undesirable central core of unmixed swirled fluid is observed throughout the combustor for all the configurations considered here.

In general, the results show increased mean axial velocities at the quench module close to the wall, due to the simultaneous contribution of the annular cooling jet and of the contraction at the exit of the rich zone. The magnitude of the peaks in the velocity distributions is, however, larger in the region between holes (Fig. 3b), which may be attributed to the blockage imparted by the quench jets which emerge with a radial velocity 25% larger than the maximum axial velocity in the

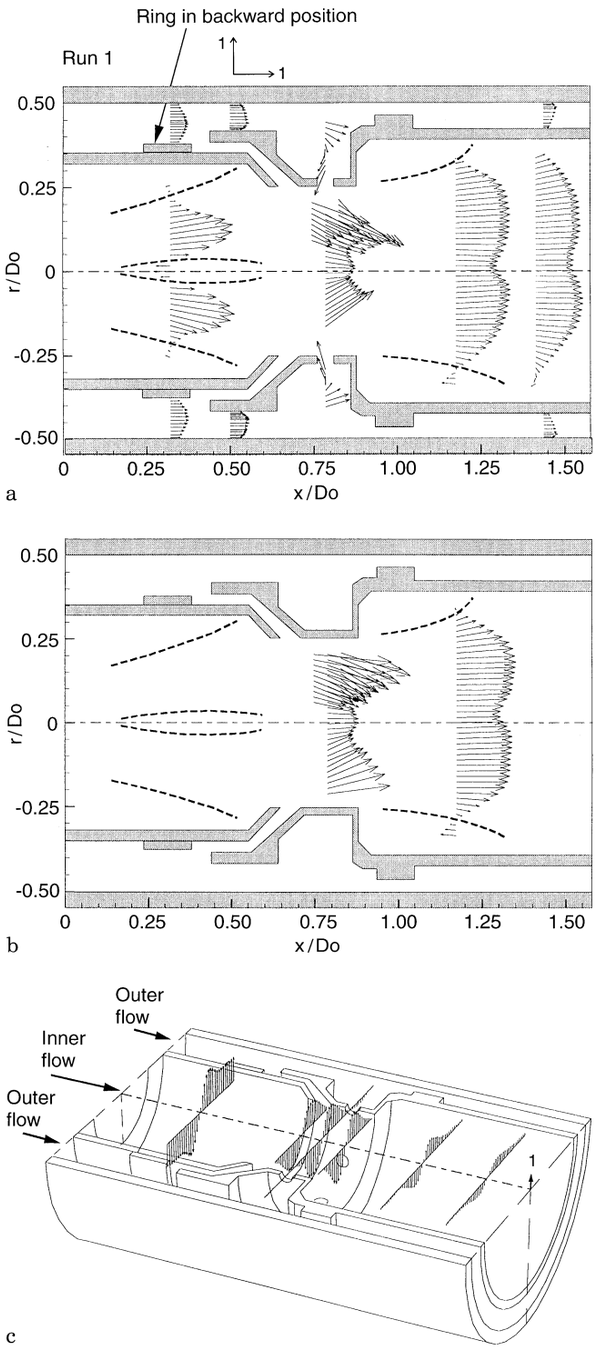


Fig. 3a–c. Mean flow patterns for Run 1. **a** Mean velocity vectors along a longitudinal plan passing through the *centre* of the quench holes (plan A in Fig. 2a) for $x/D_0 = 0.32$; 0.745; 0.786; 0.826; 1.174; 1.414; **b** mean velocity vectors along a longitudinal plan passing between two consecutive quench holes (plan B in Fig. 2a) for $x/D_0 = 0.435$; 0.786; 0.826; 1.174; **c** transversal velocity distribution related with the vectors of **a**

primary flow. A recirculation zone is formed in the wake of the jets, close to the wall, where negative axial velocities are of the same order of those in the swirl-induced recirculating zone. Also, the distributions of the tangential mean velocity show similar values, whether upstream and downstream of the quench holes, suggesting that the quench jets are not strong enough to disturb the tangential velocity of the primary flow. The core fluid remains in solid body rotation within the quench module, and the simultaneous effect of the contraction and of the annular cooling jet is mainly to alter the rotating motion of the annular fluid, closer to the wall, to a pattern typical of a free vortex.

The results also show that the sudden expansion downstream of the quench zone fulfill the need to increase the residence time in the lean zone of practical combustors, in order to promote the oxidation of the unburned products generated at the rich zone. Nevertheless, an outer annular recirculation zone is found close to the wall, which may increase combustor wall temperature.

In addition, the results show that the swirling motion is not eliminated up to the exit section of the combustor, as one could expect from the type of flow in the quench module where mainly radial momentum is added to the flow. It should be noted that the strength of swirl along plane B is similar to that in plane A (as shown in Fig. 3c) and is not shown here for lack of space. Similar aspects have been reported in other model combustors (e.g. Koutmos and McGuirk 1989), but the related flow pattern should be clearly avoided in any combustor design because it will give rise to high concentrations of unburned products in the central core of the flow at the exit of the combustor section.

Figure 4 shows radial profiles of the axial, tangential and radial fluctuating velocity components for Run 1 measured at the rich, quench and the lean zones of the combustor and characterise the turbulence levels typical of RQL combustors. The central flow in the primary zone is surrounded by a shear layer with large positive gradients of the mean axial and tangential velocities, where maximum turbulent fluctuations occur with local values up to $\sqrt{\overline{u'^2}} = 1.9\,U_0$. These values are comparable with other published in the literature (e.g. Koutmos and McGuirk 1989; Palma and McGuirk 1995), and suggest the likely representativity of the present flow pattern. The results are typical of high strained recirculating flows with swirl (see Sislian and Cusworth 1986; Heitor and Moreira 1992 for reference), in that the distribution of velocity fluctuations peak in the inner shear layer where the mean shear strains $\partial\bar{U}/\partial r$ and $\partial\bar{W}/\partial r$ are positive, but do not peak in the outer shear layer, suggesting the likely contribution of streamline curvature to turbulent production and transport. Although the detailed analysis of the turbulent flow requires plotting the results in streamline coordinates, Fig. 4 shows also that the magnitude of the axial velocity fluctuations in the primary zone is larger than those in the radial and tangential directions, with maxima differences up to 40%. In spite of the lack of optical access, which precluded measurements to be obtained in the stagnation region, it is likely that the radial velocity fluctuations increase further downstream, as observed in other recirculating flows with a free stagnation point (e.g. Taylor and Whitelaw 1984), because the relative magnitude of the three fluctuating velocity components is altered downstream of the quench module. These features suggest the likely importance of the generation of turbulent kinetic energy through the interaction between normal strains and normal stresses and are important to be noted because they imply that calculations

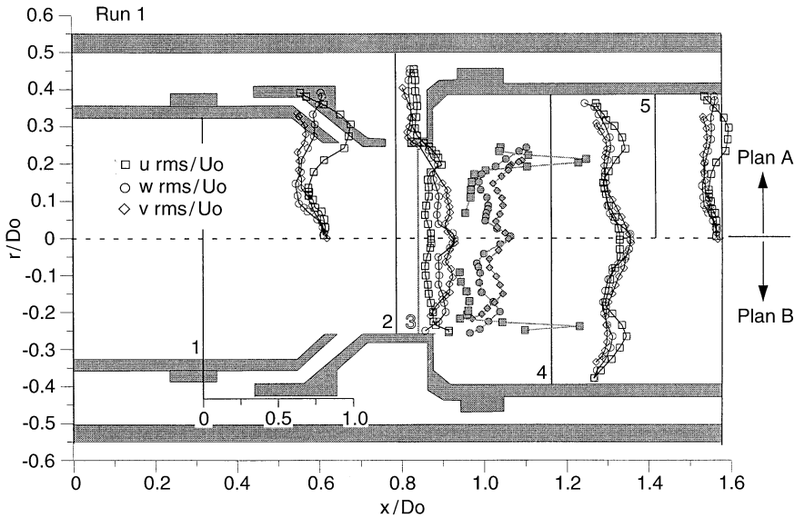


Fig. 4. Radial profiles of the axial, tangential and radial fluctuating velocity components for Run 1 along the model combustor, for $x/D_0 = 0.32$; 0.786; 0.826; 1.174; 1.414

using a scalar effective-viscosity turbulence model (e.g. Coupland and Priddin 1987; McGuirk and Palma 1992) will be inaccurate, at least for the turbulence upstream of the exit of the combustor.

The velocity fluctuations in the quench zone peak in the highly strained shear layer surrounding the quench flow and show larger values in the axial direction in accordance with the generally accepted mechanism of production of turbulent kinetic energy by the interaction between turbulent motion and the mean rate of shear strain. As noted before, the tangential and radial velocity fluctuations in the central part of the flow have local maximum with values about twice larger than those of $\sqrt{u'^2}$. Similar trends have been reported by Green and Whitelaw (1983), and Koutmos and McGuirk (1989) and have been attributed by Bicen et al. (1989) to the impingement of opposed jets at the centreline. Nevertheless, the same trend has also been observed downstream of a free stagnation point and it is likely that this mechanism holds in the present case. Further downstream, although $\sqrt{v'^2}$ and $\sqrt{w'^2}$ have local maximum around the centreline, they are of the order of magnitude of those of $\sqrt{u'^2}$, suggesting the important role of turbulent diffusion and dissipation.

The anisotropy of the turbulent flow observed so far in some zones of the flow may raise the question of whether velocity fluctuations are associated with unsteady mechanisms, such as precessing of the vortex core or vortex shedding in highly strained regions. The existence of such mechanisms would not only affect the performance of the combustor, but also would invalidate any attempt to simulate the flow with numerical models based on steady equations (McGuirk and Palma 1995b). In order to address this question, the probability density functions of velocity fluctuations and the related spectra were analysed and sample results are presented in Fig. 5. Clear bimodal probability distributions were not observed throughout the measurable flow, suggesting that instabilities, if present, do not contain enough energy to dominate over the turbulence generated fluctuations. The distribution of skewness and flatness showed axisymmetric patterns with trends in accordance with gradient transport assumptions and with values ranging within [−1, 1] and [2.5, 4.5], respectively. It is worth mentioning that the skewness and the flatness values do not vary along the azimuthal direction, even in the vicinity of the quenching jets, suggesting the likely contribution of swirl to decrease the azimuthal gradients in the quench module.

These results are to be contrasted with those recently published by McGuirk and Palma (1995a, b) in a model of a can-type gas turbine combustor, where bimodal PDFs of the azimuthal velocity distributions were observed with a broad range of frequencies (10–150 Hz), due to the jet instabilities. It was shown that imparting the swirl to the primary flow improved stability but, above a certain level, it gave rise to precessing of the vortex core along the combustor centreline.

The present spectra of velocity fluctuations in regions of large anisotropy do not show any evidence of flow periodicity, except at the edge of the annular reverse flow set up in the rich module, where a sharp peak around 10 Hz is observed. This peak is associated with skewness and flatness values around 0.5 and 3, respectively. If a Strouthal number is defined as fL/U, where f is the measured frequency, L is the width of the expansion (30 mm) and $U = 2$ m/s is the mean velocity at the edge of the step, a value of about 0.15 is obtained, which is close to the values reported by other authors in the wake of bidimensional obstacles (e.g., Durão et al. 1988).

4.2 Parametric analysis

In terms of the performance of a practical combustor, the results obtained for the standard case and discussed in the previous paragraphs show that the mixing pattern downstream of the primary zone is poor and may limit the application of the technology studied here. Possible corrections to the combustor design include the following aspects: i) elongate the quench jets, so that they impinge over a larger zone of the primary flow; ii) decrease the mass flow rate through the swirler and add crossing jets in the lean zone, in order to decrease the

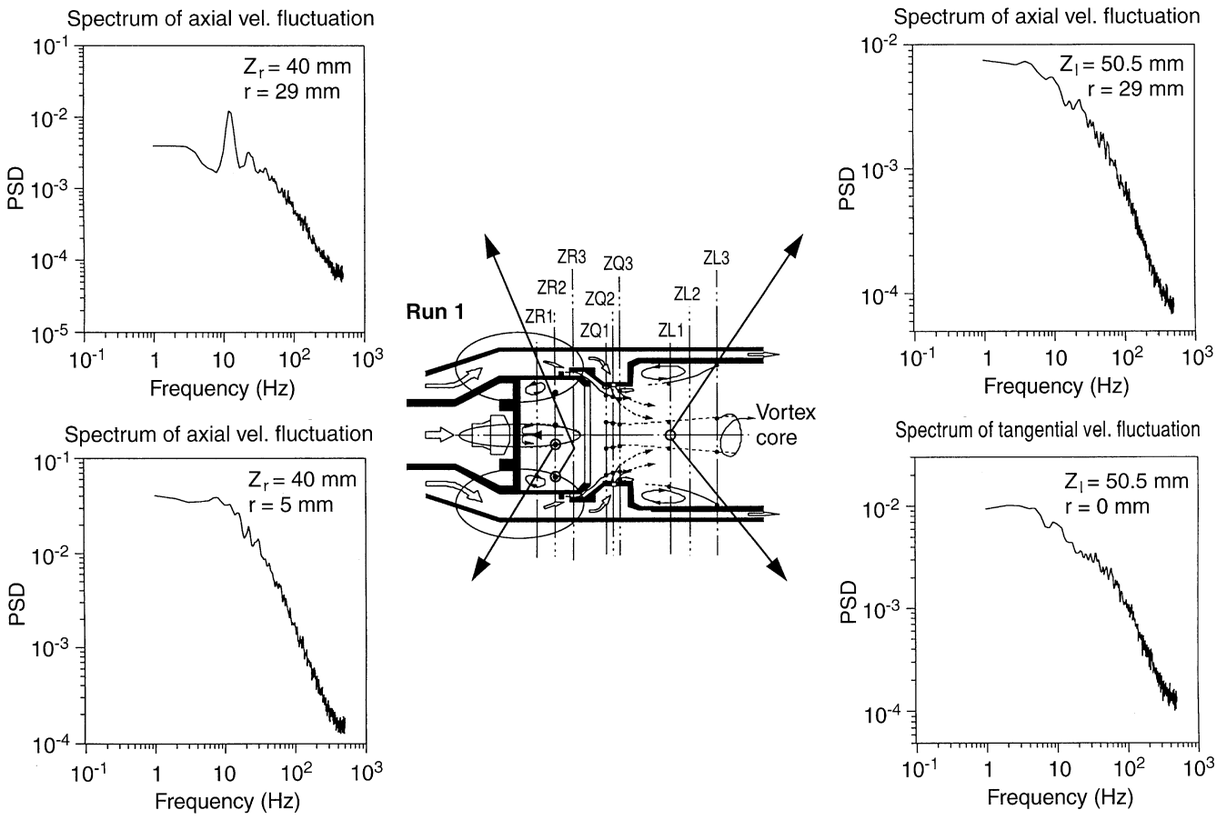


Fig. 5. Sample spectra of axial and tangential velocity fluctuations obtained in regions of large turbulence fluctuations

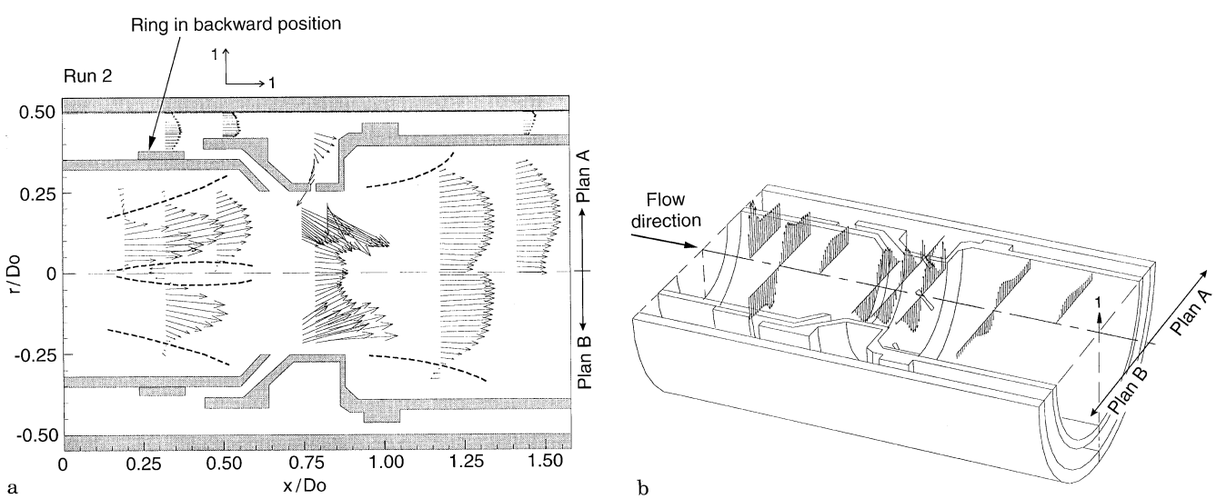


Fig. 6a, b. Mean flow pattern for Run 2. **a** Mean velocity vectors along the longitudinal plans A and B for $x/D_0 = 0.192$; 0.320; 0.480; 0.745; 0.786; 0.826; 1.174; 1.414; **b** transversal velocity distribution related with the vectors of **a**

angular momentum at the combustor outlet; iii) close the annular cooling jet at the entrance of the quench module, in order to avoid the deflection of the quench jets. These three solutions correspond to Runs 2, 3 and 4 of Tables 1 and 2 and are analysed in the following paragraphs in terms of their effects on the performance of the combustor.

i) Analysis of the influence of the shape of the quench holes – Run 2

Run 2 is characterised by the operating conditions of Run 1, but with the combustor operated with oblong quench holes, although with an area similar to the circular holes of Run 1. Figures 6 and 7 show the related mean and fluctuating velocity

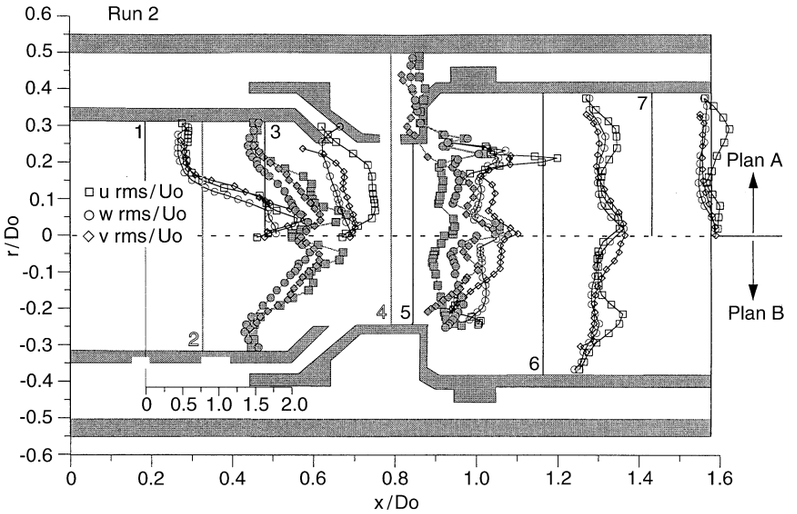


Fig. 7. Radial profiles of axial, tangential and radial fluctuating velocity components for Run 2 along the model combustor, for $x/D_0 = 0.192$; 0.320; 0.480; 0.786; 0.826; 1.174; 1.414

distributions throughout the model combustor. The measurements included analysis of three different stations in the primary zone and, in general, the results show flow characteristics similar to those described before, without any major influence of the shape of the quench holes on the mean and turbulent flow patterns. The unique exception is concerned with the profiles of tangential velocity, which now displays a significant peak close to the entrance of the quench jets. This is attributed to the shape and inclination of the quench holes, which favour the exchange of angular momentum with the primary flow.

The turbulent flow, Fig. 7, is affected in accordance with the changes observed in the mean velocity profiles. Nevertheless, it should be noted that in the vicinity of the centreline, the radial velocity fluctuations increase by about 50%, probably due to a higher relative intensity of the vortex core which is associated with the smaller penetration of the quench jets.

The relative alterations in the flow observed in the quench module are suppressed further downstream by turbulent diffusion. The mean and the turbulent fields in the lean zone show quantitative features similar to those observed in the standard case, in that the oblong jets do not attenuate the swirl motion at the outlet of the combustor.

ii) Analysis of the influence of the flow split on the flow pattern

Run 3 was achieved by decreasing the rate of primary flow to the total mass flow from 24% to 15% and making use of four holes with 8 mm in diameter in the lean zone, 49 mm downstream of the sudden expansion, in order to suppress the swirl motion at the outlet of the combustor. The ring mounted at the external surface of the rich module was moved to a forward position, which increased the blockage at the inlet port of the annular cooling jet. As a consequence, the flow rate of the annular cooling jet decreased from 39% to 31% and the flow rate of the quench jets increased from 11 to 13% of the total mass flow rate. For these conditions, the flow rate through the four dilution jets is about 8% of the total mass flow rate.

Figures 8 and 9 show sample results which characterise the mean and fluctuating velocities measured in the rich, quench

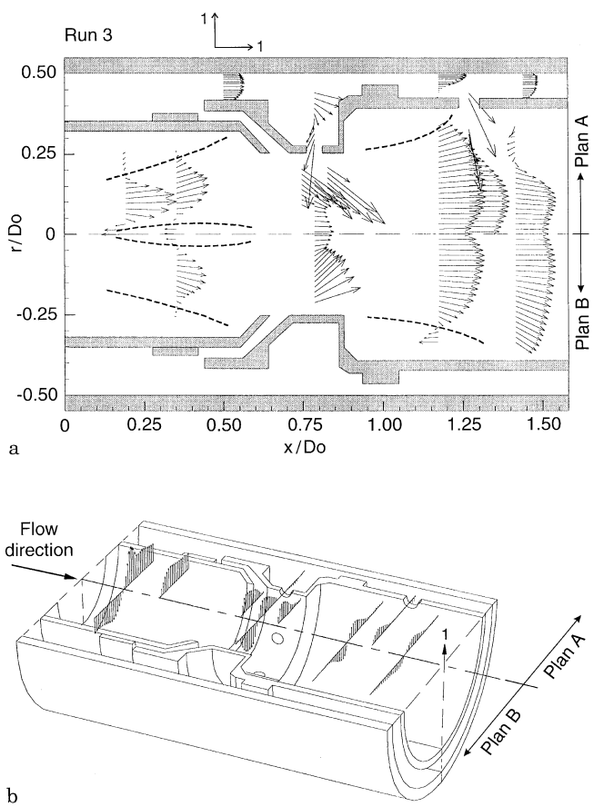


Fig. 8a, b. Mean flow patterns for Run 3. **a** Mean velocity vectors along the longitudinal plans A and B for $x/D_0 = 0.192$; 0.352; 0.745; 0.786; 0.826; 1.174; 1.270; 1.414; **b** transversal velocity distribution related with the vectors of **a**

and lean zones of the combustor. The results in the rich zone show a flow pattern similar to that measured for the standard case with maximum values of the axial and tangential mean velocities smaller by about 48% at $x/D_0 = 0.352$ (second measuring station in the rich zone), although the flow rate

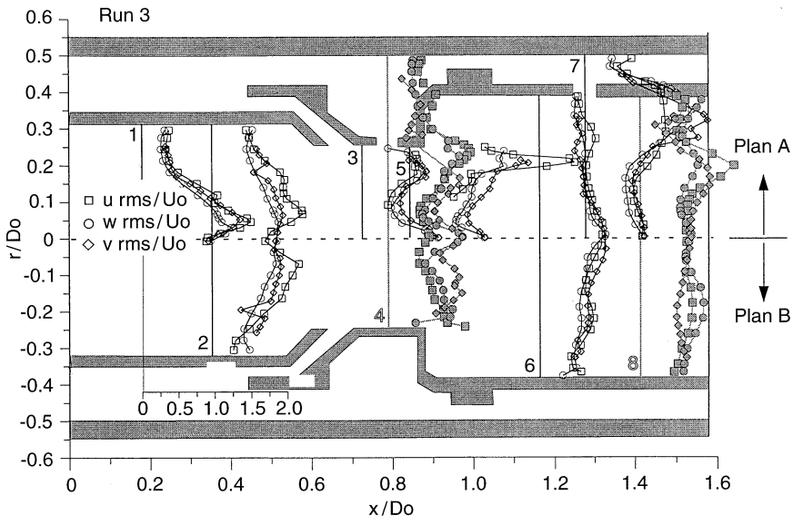


Fig. 9. Radial profiles of axial, tangential and radial fluctuating velocity components for Run 3 along the model combustor, for $x/D_0 = 0.192$; 0.352; 0.745; 0.786; 0.826; 1.174; 1.270; 1.414

through the swirler decreased by only 30%. This is associated with the different shapes and intensities of the central recirculation zones formed for the two flow conditions, which has increased in width by about 50% from Run 1 to Run 3. Although the penetration of the quench jets has increased in the present configuration, they do not generate any back flow towards the primary zone as in McGuirk and Palma (1995). The results quantify a different pattern in the quench module relative to that described before, in that immediately upstream of the quench holes (first station in the quench module in Fig. 8) the mean radial velocity is zero around the centreline, as a result of the comparatively smaller pressure forces exerted by the contraction. The magnitude of the mean tangential velocity in the vicinity of the wall is now very small, which may favour the penetration of the quench jets. However, and despite these jets now emerge with a radial velocity 20% larger than in the standard case, the hump-like profile of the mean axial velocity measured further downstream in the lean zone, show that the jets still do not impinge at the centreline. Also, the peak values of the axial mean velocity close to the wall remain similar to those measured in the standard case, suggesting the likely importance of the annular cooling jet.

For the analysis of the flow in the dilution zone, it should be noted that since the dilution holes are aligned with the quench holes but they are half in number, the plan A of the measurements crosses the centres of the dilution holes, but the plan B does not cross at half distance between two consecutive holes. The results show that the use of holes in the lean zone gives rise to a new set of small recirculation zones in the wake of the related jets, which emerge with a radial velocity similar to that of the quench jets. The dilution jets penetrate through the flow within a radial distance similar to that of the quench jets, although the velocity fluctuations in their wake are relatively smaller and with considerably low values along plane B. This may be associated with the large distance between two consecutive holes and the large diameter of the lean zone, which give rise to smaller velocity gradients in the flow region close to the entrance of the jets.

Analysis shows that the tangential momentum decreases effectively from the quench to the lean zones in contrast to the standard case, which suggests the beneficial effect of the dilution jets. However, this is brought together with a reduction of the expected fuel/air ratio in the lean zone of practical combustors, which may limit the present concept. Practical operation may also have to consider other aspects, such as the onset of aerodynamic induced oscillations associate with the coupling between the various sets of jets.

iii) Analysis of the influence of the annular cooling jet strength on the flow

The analysis above emphasises the importance of the annular cooling jet in deflecting the quench jets, thus precluding their penetration through the primary flow. To allow to complete the analysis, Run 4 was considered with the inlet port of the cooling jet closed. Although this may not be suitable for practical combustors, it provides a better understanding of the mixing pattern within the quench module. As a consequence, the flow split at the entrance of the model remained similar to the standard case, but the flow rate of the quench jets increased by about 72% and those of the dilution jets by about 115% (see Table 2).

Figure 10 and 11 characterise the main features of the mean and turbulent flows associated with Run 4, which show a primary flow similar to that of Run 1, thus confirming the dominant role of swirl in the upstream flow. Nevertheless, the distribution of the axial mean velocity measured further downstream shows that the swirl induced recirculation zone is now forced by the quench jets to stagnate at the plane of the holes. The quench jets emerge with a radial velocity about six times larger than the maximum axial velocity of the primary flow and induce large azimuthal gradients of mean and fluctuating velocities. In addition, the impingement of opposed jets in the centre of the combustor enhances turbulent production, giving rise to velocity fluctuations about five times larger than in the standard case. This is clearly observed either in the shear layer where turbulent fluctuations are mainly due to shear induced mechanisms, or in the vicinity of the

centreline where the contribution of normal strains to turbulence production is expected to be important (e.g. Heitor and Whitelaw 1986).

For the present flow conditions, the primary flow interacts with a smaller mass rate of non-swirling fluid and the quench jets are only able to deflect the peaks of the tangential mean velocity towards the centreline. As a result, the core fluid in the quench module now rotates with an angular velocity about twice larger than in the standard case. Consequently, the axial mean velocity exhibits a single peak at the centreline due to the absence of the annular cooling jet and the angular motion diffuses radially outwards, giving rise to a solid body rotation pattern over all the cross section of the combustor. The fluid leaving the lean zone interacts further downstream with the annular row of dilution jets, which reduce the swirl motion at the outlet of the model in terms of absolute flow rates.

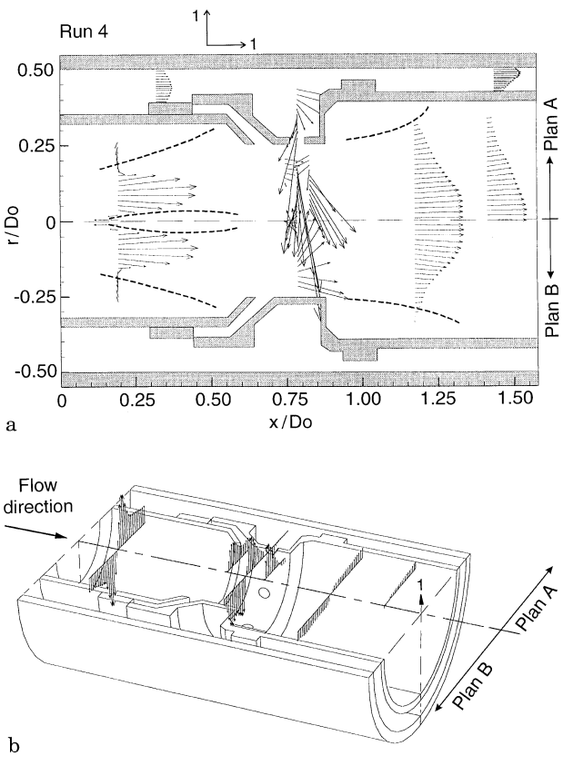


Fig. 10a, b. Mean flow patterns for Run 4. **a** Mean velocity vectors along the longitudinal plans A and B for $x/D_0 = 0.192$; 0.745; 0.786; 0.826; 1.174; 1.414; **b** transversal velocity distribution related with the vectors of **a**

5 Conclusions

The flow in a RQL tubular gas-turbine combustor was studied based on detailed laser-Doppler measurements of the velocity field in a water model. The study includes analysis of a basic flow configuration and a series of geometrical modifications in order to study the influence of different parameters on the performance of the combustor, such as the swirl level in the rich zone, the geometry of the quench holes, and the momentum of the inflowing jets and of the annular cooling jet on the internal flow field. The flows studied encompassed Reynolds numbers based on the bulk axial velocity at the outlet of the model, about 10^5, which are representative of real combustor flows.

The work has been derived from the need to extend knowledge of preliminary full-scale combusting experiments, which have been shown to be limited in extent, time and the skills required. The prime objectives are to improve understanding of major flow features and of the effect of the details of the combustor geometry on the flow pattern, as well as to provide the basis for the numerical calculation of the flow in this type of combustors.

Detailed measurements of the three mean and fluctuating velocity components of the flow show that the performance of the combustor is strongly influenced by the mixing rates at the quench stage and by the momentum transfer throughout the combustor. The main features of the flow structure are as follows: i) a swirl driven recirculation zone at the rich zone, which is surrounded by a shear layer characterised by strong velocity fluctuations and limited in extent by the flow in the

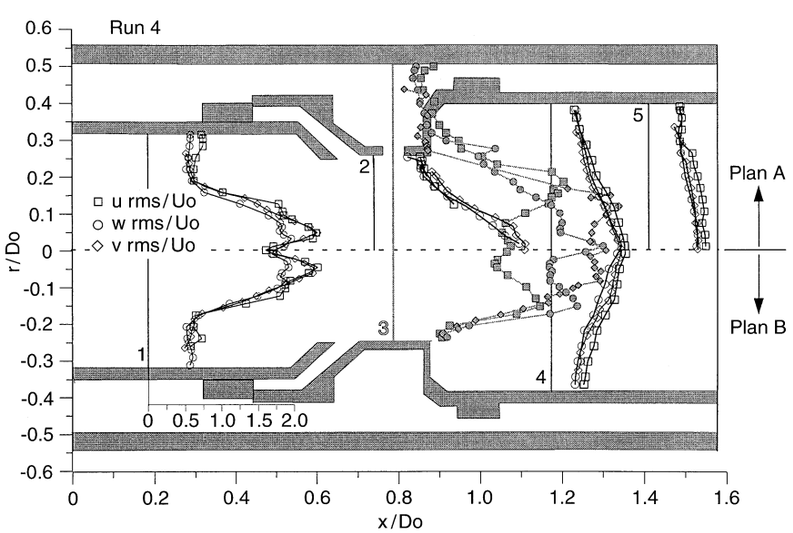


Fig. 11. Radial profiles of axial, tangential and radial fluctuating velocity components for Run 4 along the model combustor for $x/D_0 = 0.192$; 0.745; 0.786; 1.174; 1.414

quench zone; ii) an outer recirculation zone of low intensity close to the wall in the rich zone, which is limited by the contraction at the entrance of the quench module and characterised by periodic motions at about 10 Hz; iii) a reduced penetration of the quench jets, which is in part limited by the impingement of the annular cooling jet; iv) a core fluid at the outlet of the lean zone, with a significant swirl component.

Other main conditions include the following aspects:

- The decrease of the angular momentum at the outlet of the lean zone can only be achieved by decreasing the flow rate of primary fluid through the swirl cup and by adding a row of crossing jets at the lean stage. In practical operation, such method may be limited by the minimum fuel/air ratio necessary for the oxidation of unburned gases in the lean stage.
- The turbulent fluctuations at the quench stage, which determine the tubulent mixing under burning operation, only depend on the flow pattern induced by the annular jet used to cool the front wall of the rich module and by the quench jets. Enhanced turbulent fluctuations can be achieved by increasing the momentum of the quench jets relatively to the axial momentum of the annular cooling jet. However, a limitation is also imposed since the pressure distribution imparted by the annular jet is essential to keep the swirl induced recirculation zone within the rich zone of the combustor and, thus, to avoid combustion reactions to propagate down to the quench stage.

The present results provide detailed information on the aerodynamic field in a practical geometry and can be used to evaluate the performance of calculation methods, which can then be used to interpolate and extrapolate the experimentally-acquired information, in order to improve the design of practical combustors. The mean flow includes zones of flow reversal, rotation, jet impingement through a cross flow and dilution, resulting in complex 3-D flow patterns. In addition, the distribution of normal stresses show substantial departures from local isotropy and their ratios appear to be strongly dependent on the local mean strain rates. In general the results suggest that it is necessary to calculate the individual normal stresses adequately, if the correct turbulent kinetic energy is to be obtained. This may preclude the use of turbulence models at the level of a turbulent viscosity closure and requires consideration of adequate mixing models.

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